Cooperative Robots for Smart Factories

An open-source logistics integration blueprint

iisys
Institut für Informationssysteme der Hochschule Hof

7th Scientific Institute Colloquium

Abstract

A TurtleBot 4 and a MyCobot 280 jointly run three logistics missions in a didactic Siemens smart factory, linked to a real-time 3D twin of the same cell. Only open-source software and onboard sensors are used.

The integration challenge

Mobile robots, manipulators and factory PLCs run on separate software stacks. ROS 2 Jazzy, ROS 2 Galactic and the Siemens S7-1500 network have to be coordinated explicitly to work as one system.

A modular solution

Three replaceable modules — mobile base, arm and OPC UA factory gateway — meet at fixed interfaces, so any one can be swapped out or run on its own.

Cooperative delivery pipeline

SLAM mapping · AMCL localisation · Nav2 navigation

MoveIt 2 motion planning + suction gripper

Perception
Camera + YOLO
Detect product
Validate class
Send class

class →

Coordination
Mission manager
Select mission
Check readiness
Set destination

task →

Pickup
TurtleBot 4
Undock
Drive to pickup
Confirm arrival

arrival →

Transfer
MyCobot 280
Wait for arrival
Place product
Confirm release

release →

Delivery
TurtleBot 4
Wait for release
Deliver by colour
Return to dock

Two handshakes keep the robots in step: the arm waits for arrival, the base waits for release.

Digital twin & factory in the loop

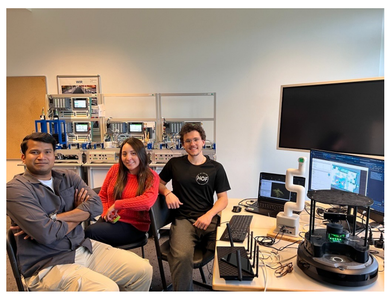
A real-time 3D twin rehearses missions on the same ROS 2 stack, and a web dashboard starts and monitors them. The factory joins as a peer: it releases the can to the arm.

Three logistics missions

Product delivery
Fetch the detected product and deliver it to the station that matches its colour.

Defective-part retrieval
Pick up rejected components at the factory and carry them to the inspection area.

Material restocking
Collect raw material, restock the factory inventory station and return to the dock.





B002 laboratory and project team.

Evaluation & outlook

Missions run on real hardware against a live Siemens S7-1500 network. ROS bag logs give cycle time and completion rate under clear, static and moving obstacles.

Supervision: Prof. Dr.-Ing. Christoph Koch
Future Lab: Shantall Marucia Cisneros Saldana

Matheus Hipólito Carvalho
EESC–USP / Hof University of Applied Sciences
matheus.carvalho@hof-university.de

Hochschule Hof
University of Applied Sciences

Institute for Information Systems
Hof University of Applied Sciences

Alfons-Goppel-Platz 1 · 95028 Hof
www.hof-university.de